(Please write your roll no. here) Exam Roll No.....

10/7/23

END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] July 2023

Paper Code- ETCS 206	**Subject: Theory of Computation**
Time: 3 Hours	**Maximum Marks:75**

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit.

Q1 Attempt all questions: **(5x5=25)**

(a) State the differnces between a NFA and DFA?
(b) Differentiate between Moore and Melay Machines?
(c) Find a regular expression corresponding to the language of all strings over the alphabet { a, b } that contain exactly two a's.
(d) State Church's hypothesis about compuitability of a machine
(e) Prove that the graph coloring problem is NP-complete

UNIT-I

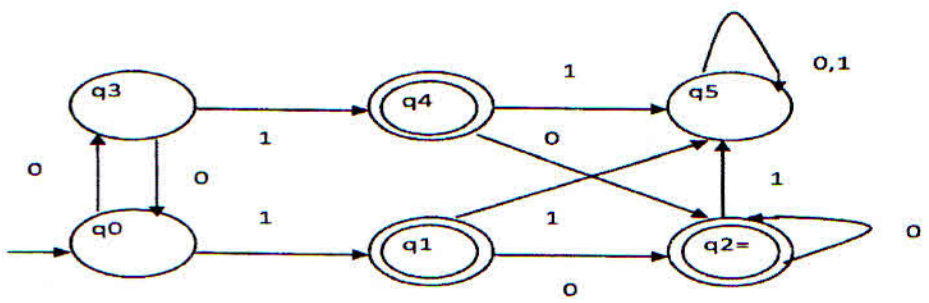
Q2 a) Let Σ = {a, b} and let L = { w $\in \Sigma^*$ | w $\neq \varepsilon$ and the first and last character of w are the same }. Design aDFA for L **(6)**

b) For the following NFA. Convert it into its equivalent DFAwhere Q0 is the initial state and Q2 is the final state. **(6.5)**

	a	b
Q0	{Q0}	{Q0,Q1}
Q1	€	{Q2}
Q2	€	€

Q3 a) Prove the pumping lemma for regular lanuages with examples. **(6)**

b) Minimize the following DFA over alphates as 0,1 **(6.5)**



UNIT-II

Q4 a) What are inherently ambiguous languages. Check whether the folllowing Grammar is Ambiguous or not-S->A/B, A->aAb /ab, B->abB/ € **(6)**

P.T.O.

P-1/2
ETCS-206